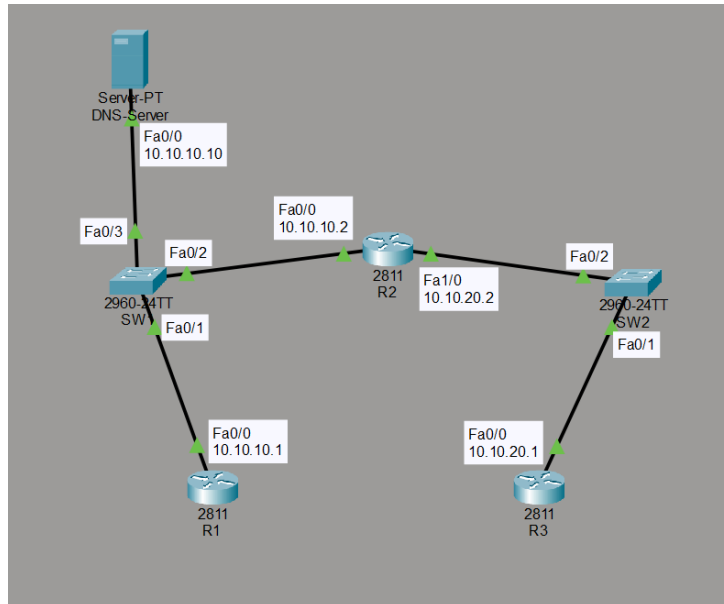


NET512 — Weekly Practical

NAME	Christiaan van Wyk	WEEK	Week 3
STUDENT NO.	20261193	DATE	16 August 2026

Topology

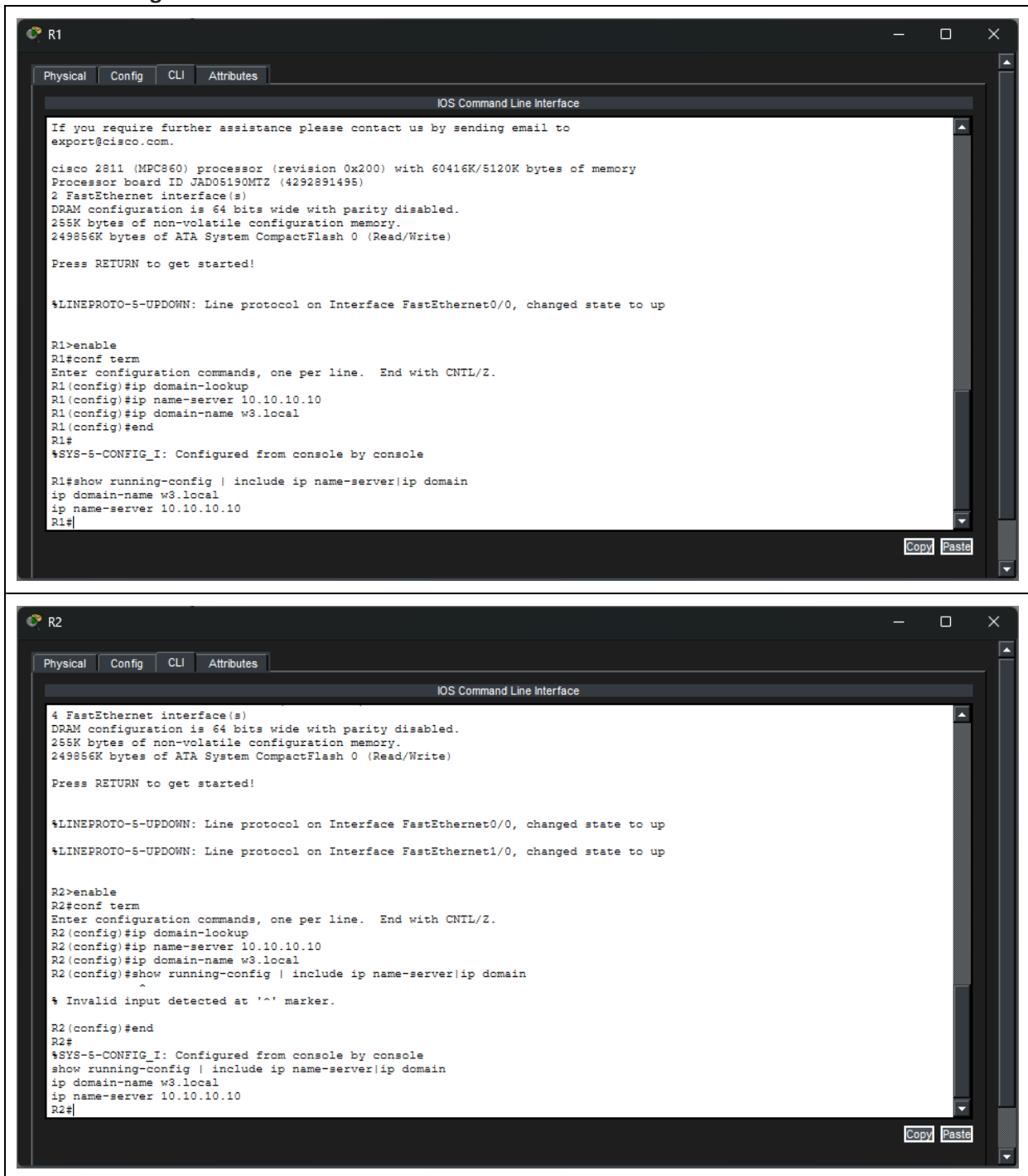


1[Baseline - Connectivity]

```
R3
Physical Config CLI Attributes
IOS Command Line Interface
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html
If you require further assistance please contact us by sending email to
export@cisco.com.
cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
Press RETURN to get started!
R3>enable
R3#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
R3#
```

Task 16 — Configure the routers as DNS clients

Client DNS Configuration



The image displays two screenshots of the Cisco IOS Command Line Interface (CLI) for routers R1 and R2, showing the configuration of DNS clients.

R1 Configuration:

```
R1
Physical Config CLI Attributes
IOS Command Line Interface
If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
2 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

R1>enable
R1#conf term
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-lookup
R1(config)#ip name-server 10.10.10.10
R1(config)#ip domain-name w3.local
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show running-config | include ip name-server|ip domain
ip domain-name w3.local
ip name-server 10.10.10.10
R1#
```

R2 Configuration:

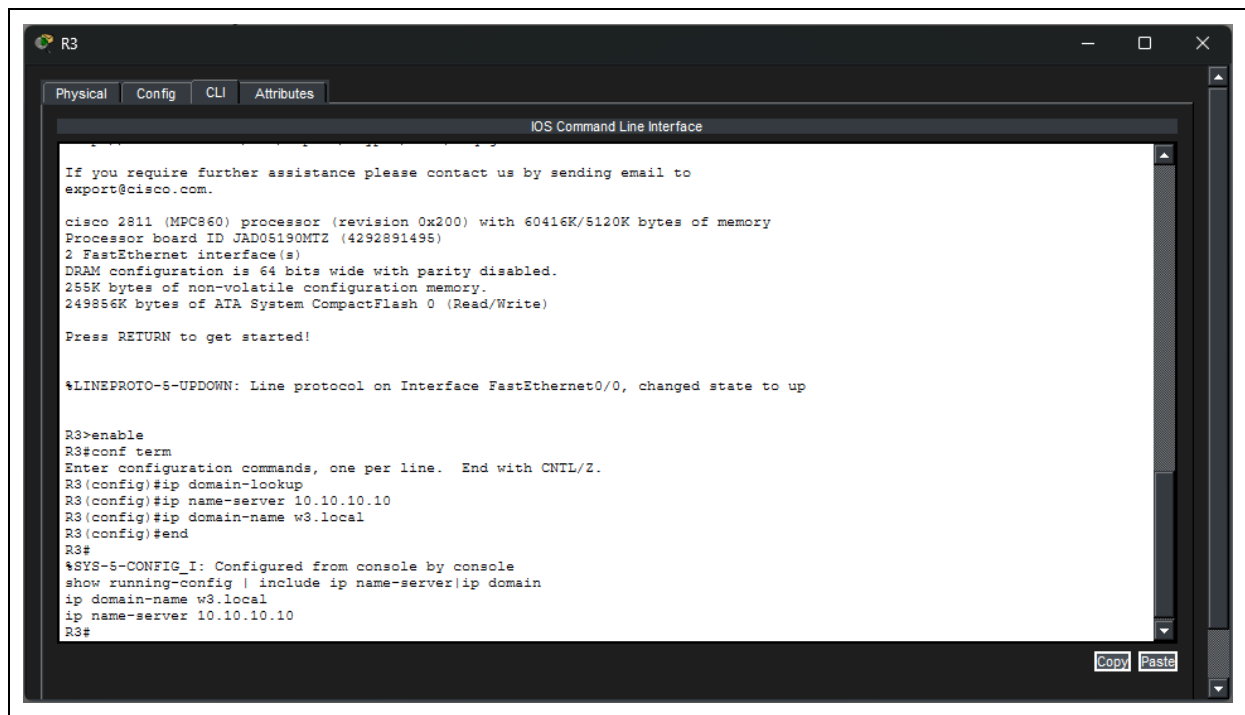
```
R2
Physical Config CLI Attributes
IOS Command Line Interface
4 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

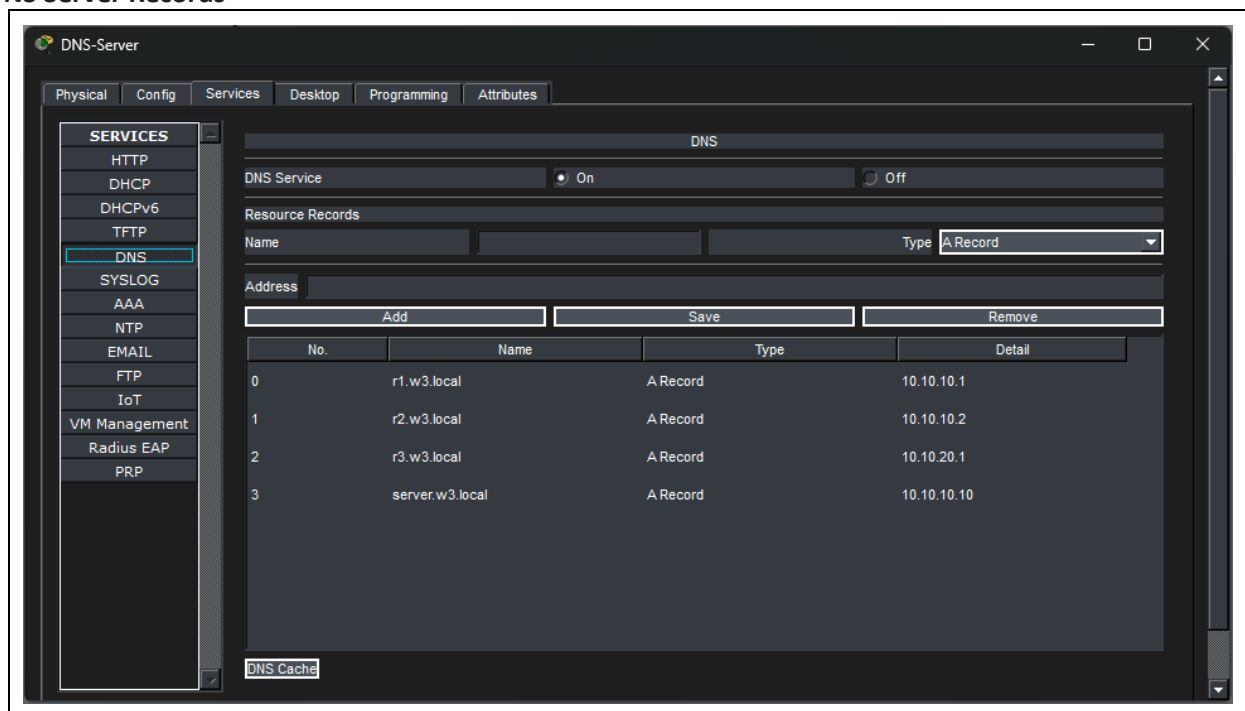
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R2>enable
R2#conf term
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip domain-lookup
R2(config)#ip name-server 10.10.10.10
R2(config)#ip domain-name w3.local
R2(config)#show running-config | include ip name-server|ip domain
^
% Invalid input detected at '^' marker.

R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
show running-config | include ip name-server|ip domain
ip domain-name w3.local
ip name-server 10.10.10.10
R2#
```

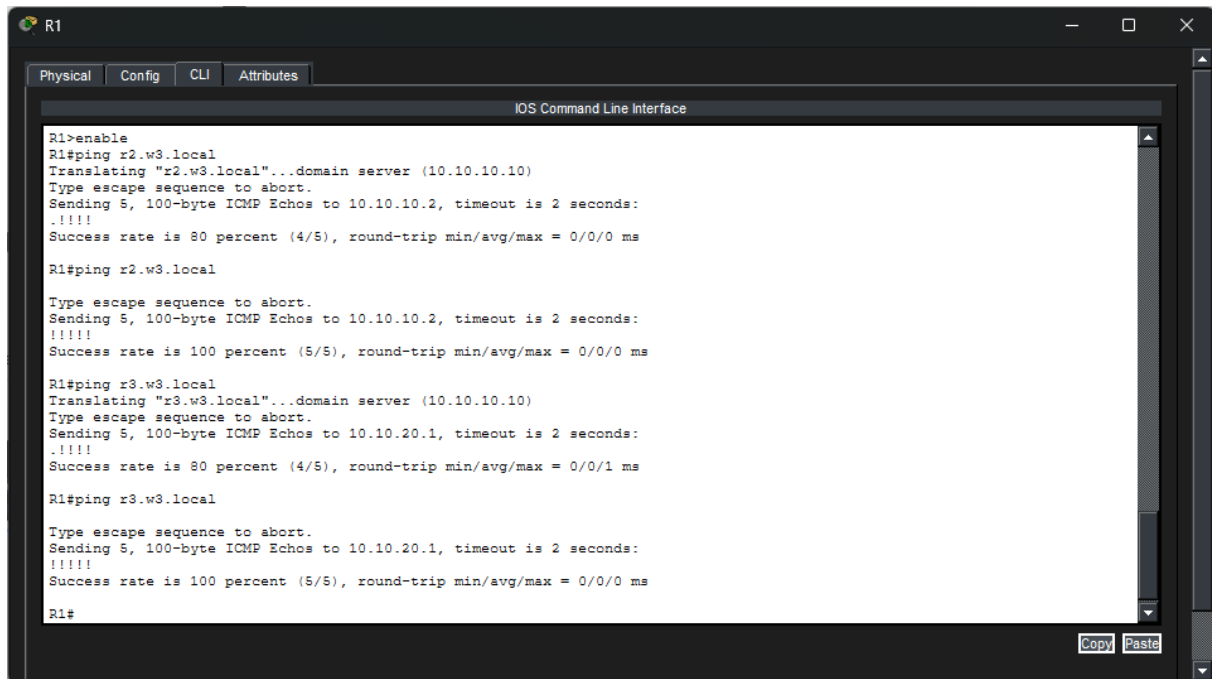


DNS Server Records



Task 17 — Test name resolution

Name Resolution



```
R1
Physical Config CLI Attributes
IOS Command Line Interface

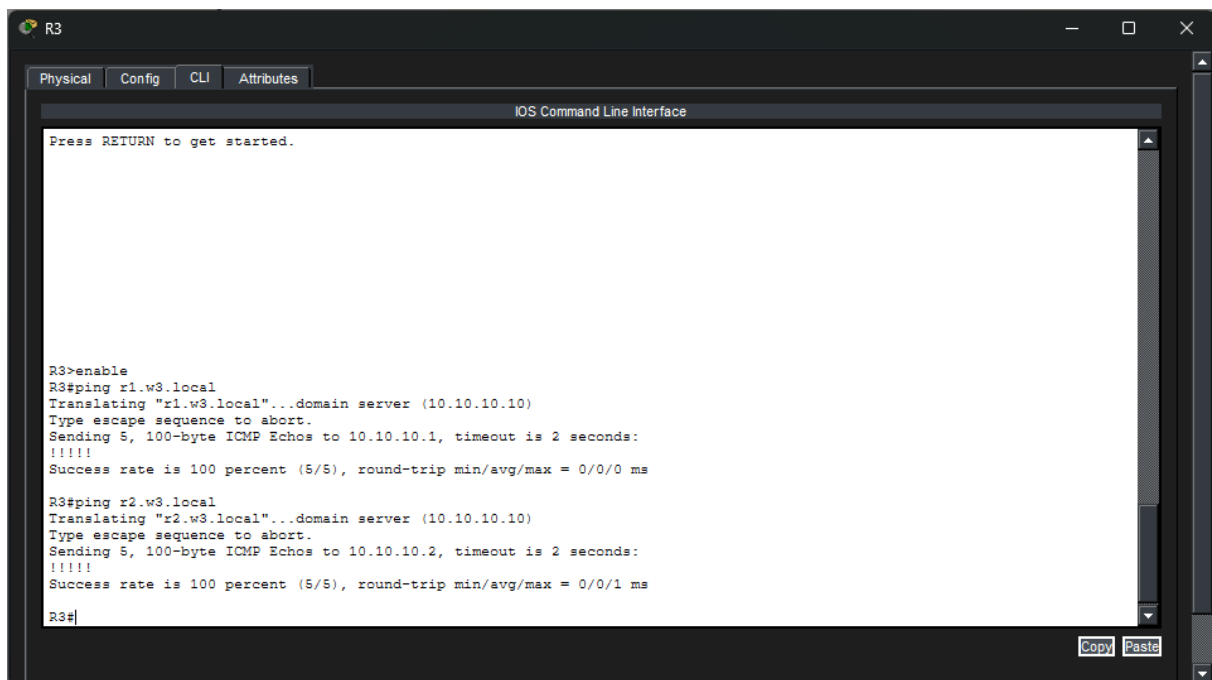
R1>enable
R1#ping r2.w3.local
Translating "r2.w3.local"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

R1#ping r2.w3.local
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#ping r3.w3.local
Translating "r3.w3.local"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms

R1#ping r3.w3.local
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```



```
R3
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

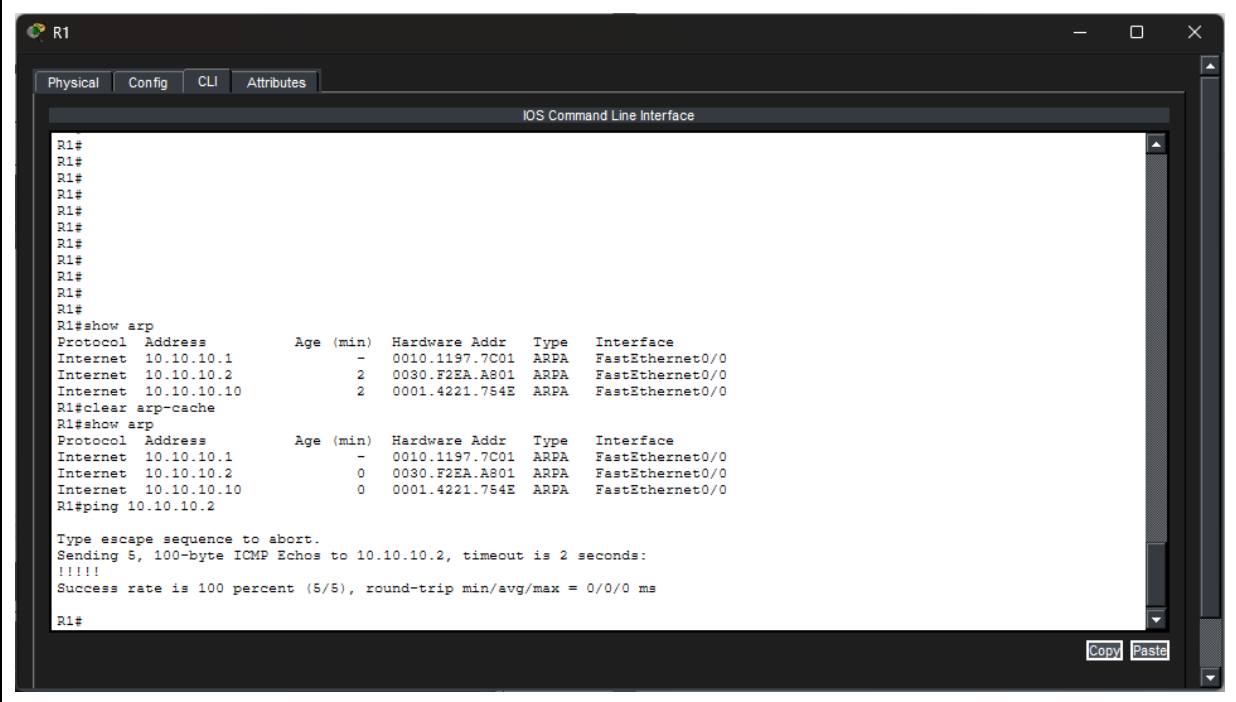
R3>enable
R3#ping r1.w3.local
Translating "r1.w3.local"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R3#ping r2.w3.local
Translating "r2.w3.local"...domain server (10.10.10.10)
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R3#
```

From the screenshots we can see that the Routers held the names but did not have the address for them. Therefore, they sent a Query to the DNS server and received the Records telling them the address.

Task 18 — Clear the ARP cache, then generate traffic



The screenshot shows a Cisco IOS Command Line Interface (CLI) window for router R1. The window has tabs for Physical, Config, CLI, and Attributes. The CLI shows the following commands and output:

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
-----
Internet 10.10.10.1          -    0010.1197.7C01  ARPA   FastEthernet0/0
Internet 10.10.10.2          2    0030.F2EA.A801  ARPA   FastEthernet0/0
Internet 10.10.10.10        2    0001.4221.754E  ARPA   FastEthernet0/0
R1#clear arp-cache
R1#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
-----
Internet 10.10.10.1          -    0010.1197.7C01  ARPA   FastEthernet0/0
Internet 10.10.10.2          0    0030.F2EA.A801  ARPA   FastEthernet0/0
Internet 10.10.10.10        0    0001.4221.754E  ARPA   FastEthernet0/0
R1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
R1#
```

Task 19 — Examine the ARP cache and identify the new entry

ARP Cache Entries

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 10.10.10.1          -    0010.1197.7C01  ARPA   FastEthernet0/0
Internet 10.10.10.2          2    0030.F2EA.A801  ARPA   FastEthernet0/0
Internet 10.10.10.10       2    0001.4221.754E  ARPA   FastEthernet0/0
R1#clear arp-cache
R1#show arp
Protocol Address      Age (min) Hardware Addr  Type   Interface
Internet 10.10.10.1          -    0010.1197.7C01  ARPA   FastEthernet0/0
Internet 10.10.10.2          0    0030.F2EA.A801  ARPA   FastEthernet0/0
Internet 10.10.10.10       0    0001.4221.754E  ARPA   FastEthernet0/0
R1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

R1#
```

Copy Paste

Confirming MAC of R2

The screenshot shows a terminal window titled "R2" with a dark background. At the top, there are four tabs: "Physical", "Config", "CLI", and "Attributes". The "CLI" tab is selected, and the title bar of the terminal window reads "IOS Command Line Interface".

The terminal output displays the following text:

```

compliance with U.S. and local country laws. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local laws, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wml/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

cisco 2811 (MPC860) processor (revision 0x200) with 60416K/5120K bytes of memory
Processor board ID JAD05190MTZ (4292891495)
4 FastEthernet interface(s)
DRAM configuration is 64 bits wide with parity disabled.
256K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

R2>enable
R2#show interfaces fa0/0 | include address is
    Hardware is Lance, address is 0030.f2ea.a801 (bia 0030.f2ea.a801)
    Internet address is 10.10.10.2/24
R2#
  
```

At the bottom right of the terminal window, there are two buttons labeled "Copy" and "Paste".

Task 20 — Reflection Question

What information does an ARP entry give you that a routing table entry does not?

ARP Entries provide the MAC address of a device of the direct network. It binds the MAC to the Layer 3 IP address of the device. The routing table is purely in regard to Layer 3. It shows the name of outgoing interfaces and the IP address of the next hop. It does not contain MAC addresses at all. The routing table purely says what direction the packet goes while ARP says the physical address to supply for the packet's destination.

Task 21 — Reflection Question

Why does a router need to perform an ARP lookup even after it already knows the correct route from its routing table?

A router needs to be capable of doing ARP lookups even if it knows the correct route derived from the routing table because there are 2 separate tasks being done and both are on different layers. The routing table tells the router what ip and interface to hop to next but in order for the packet to actually be sent the router needs to encapsulate it in an ethernet frame which requires a destination MAC address as one of its fields to make sure the packet is going to the right device. ARP is only for finding the address of the next hop not for the physical hardware address.